

Roll No. :

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B.Tech. CSE**

Semester: V

Name of the Course: Foundation of Quantum Computing

Course Code: TCS 584

Time: 90 Minutes

Maximum Marks: 50

Note:

- Answer all the questions by choosing any one of the sub questions
- Each questions carries 10 marks

Q1	(10 Marks)	
(a)	Given an arbitrary single-qubit unitary gate U , show that it can be expressed as $U = e^{i\alpha}R_z(\beta)R_y(\gamma)R_z(\delta)$ for some real $\alpha, \beta, \gamma, \delta$.	CO1
	OR	
(b)	Show that multiplying a quantum state by a global phase $e^{i\phi}$ has no physical meaning and give an example of a single-qubit gate that introduces a relative phase and explain its effect on measurement outcomes.	
Q2	(10 Marks)	
(a)	Two qubits are prepared in the state $ \psi\rangle = \frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$. According to Postulate 5 (composite systems), explain why this is not a product state and if we measure the first qubit, what outcomes are possible and with what probabilities?	CO1
	OR	
(b)	If the system is in state $ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$, and observable is Pauli-Z ($Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$), show that the expectation value is $\langle Z \rangle = \alpha ^2 - \beta ^2$. Explain how this connects Postulates 2 and 3.	
Q3	(10 Marks)	
(a)	Explain that any pure qubit state lies on the surface of the Bloch sphere and show that a mixed state lies inside the sphere.	CO2
	OR	
(b)	The system is in state $ \psi\rangle = \frac{1}{\sqrt{2}} 0\rangle + \frac{1}{\sqrt{2}} 1\rangle$. If measurement is made on a computational basis, what are the possible outcomes and probabilities? After observing outcome $ 0\rangle$, what is the new state of the system? And repeat for outcome $ 1\rangle$.	
Q4	(10 Marks)	
(a)	Explain why classical states (bits) can be copied freely. Why does quantum mechanics forbid cloning of arbitrary quantum states? Why is it still possible to copy orthogonal states (like $ 0\rangle$ and $ 1\rangle$)?	CO2
	OR	
(b)	A qubit is in the mixed state $\rho = \frac{3}{4} 0\rangle\langle 0 + \frac{1}{4} 1\rangle\langle 1 $. Find the expectation values of Z, X, Y .	
Q5	(10 Marks)	
(a)	Explain (with example) the key quantum phenomenon-Superposition, Entanglement and Interference.	CO1
	OR	

(b)	Explain (with example) why projective measurement is different from classical sampling (Quantum vs Classical Measurement).	CO3
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